higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION – SHEET METAL WORK
NQF LEVEL 3

11 MARCH 2019

This marking guideline consists of 7 pages.

Copyright reserved

QUESTION 1

1.1 1.1.1
- Cast iron
- Wrought iron
- Low-carbon steel
- Mild steel
- Medium-carbon steel
- High-carbon steel (Any 1 × 1)

1.1.2
- Aluminium
- Copper
- Brass
- Bronze
- Tin
- Lead
- Zinc
- Antimony (Any 1 × 1)

1.1.3
- Manganese steel
- Nickel steel
- Chromium steel
- Nickel-chromium steel
- Stainless steel
- Vanadium steel (Any 1 × 1)

(3 × 1) (3)

1.2 1.2.1 B
1.2.2 A
1.2.3 D
1.2.4 C
1.2.5 E

(5 × 1) (5)

1.3 **NOTE: ONE mark for name of each tool and ONE mark for 1 use of each.**

1.3.1 Flat screwdriver
- To turn screws
- Tighten screws
- Loosen screws

1.3.2 Waterpump pliers
- To hold objects with smooth surfaces
- Generally used to tighten or loosen nuts in places where it is difficult to gain access with normal spanners

1.3.3 Ring spanner
- To hold or turn bolts and nuts

1.3.4 Adjustable spanner/Shifting spanner
- To hold or turn a variety of different sizes of bolts or nuts
- To fit odd-sized nuts

1.3.5 Allen key
- To hold or turn screws and bolts with hexagonal slots

1.3.6 Flat chisel
- To cut excess material off flat surfaces
- Remove rivet heads
- Cut metal sheets

1.3.7 Diamond point chisel
- To finish corners or to cut V-grooves
- Chip through plates
- Clean corners

1.3.8 Portable electric drill
- To do general drilling work

1.3.9 Compressor
- It is a source for compressed air

(9 × 2) (18)

1.4 1.4.1 Inside callipers
- To take inside measurements of round shafts

1.4.2 Outside callipers
- To take outside measurements of round shafts

(2 × 2) (4)

1.5 1.5.1 Square head
1.5.2 Protractor
1.5.3 Centre head

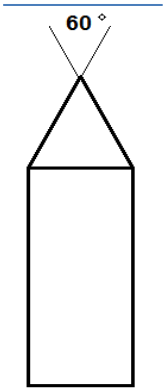
(3 × 1) (3)

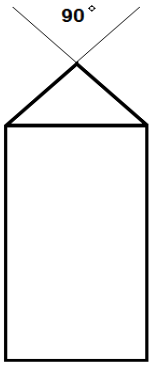
1.6 1.6.1 The square head measures the depth of a hole.
1.6.2 The centre head determines the centre of a round object.

(2 × 2) (4)

Copyright reserved

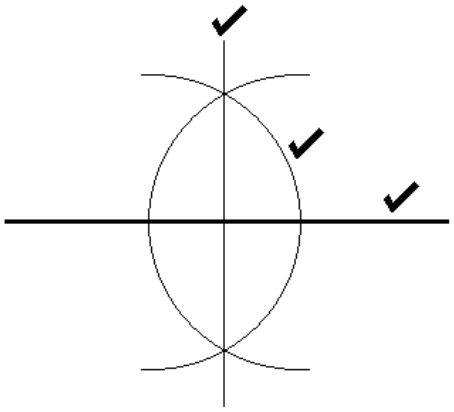
1.7





Prick punch Centre punch (2 × 2) (4)

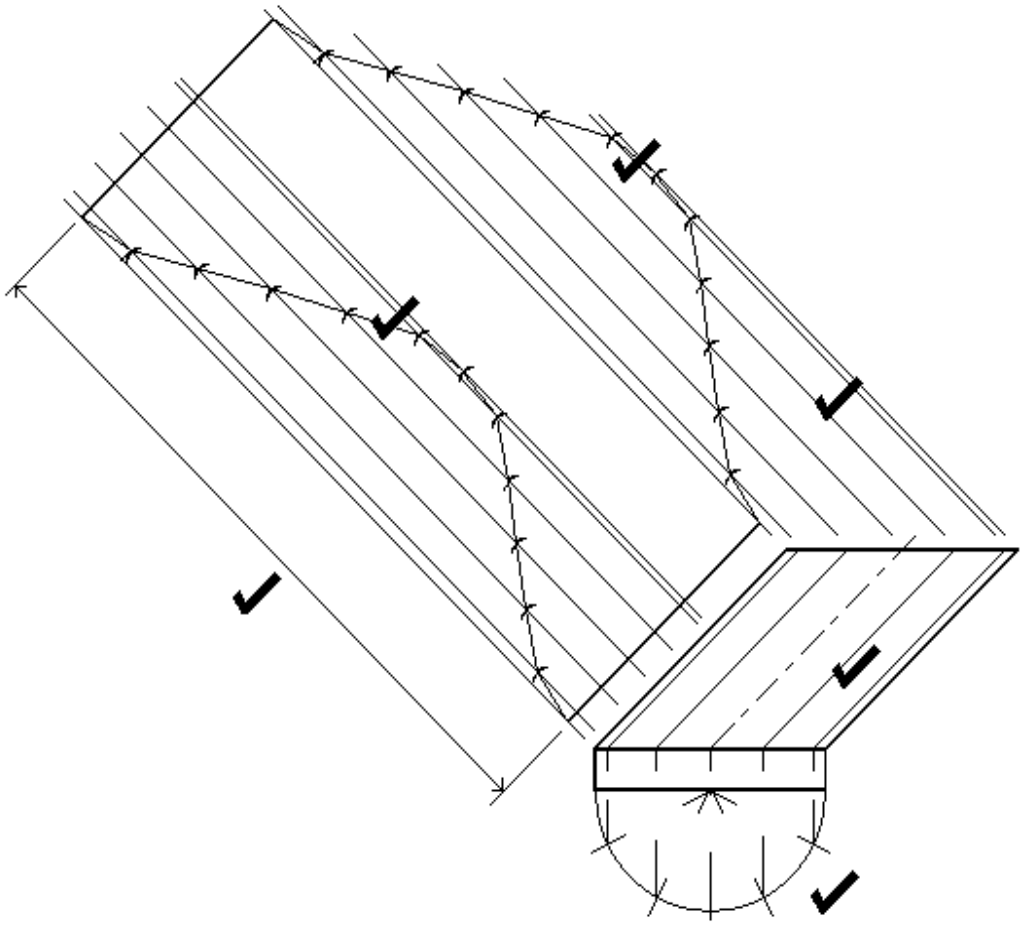
1.8



(3 × 1) (3)

1.9 Circumference = π × D (1)

1.10 **ANSWER SHEET**



MARK SHEET

CRITERIA	POSSIBLE MARK	MARK OBTAINED
Construction of circle with divisions	1	
Construction of divisions diagonally from bottom of pipe to top of pipe	1	
Construction of development lines	1	
Correct method of divisions (oblique method) and curves	2	
Correct circumference used	1	
TOTAL	**(6)**	

(6 × 1) (6)

1.11 $a^2 = b^2 + c^2$
$c^2 = a^2 - b^2$ ✔
$c = \sqrt{5,385^2 - 5^2}$ ✔
$c = 3,998$ m ✔

Thus: Total area = l × b ✔
= 5 × 3,998 ✔
= 19,990 m^2 ✔ (6 × 1) (6)

1.12 An isometric drawing shows one combined view of an object. (2)

1.13 Square notch (1)
[60]

QUESTION 2

2.1
- The welder must be trained to use the machine.
- Welders must be provided with PPE.
- Cables must be well-insulated.
- The work area must be well-screened off.
- Ventilation must be effective.
- Equipment must be well-insulated in damp areas.

(Any relevant answer) (Any 4 × 1) (4)

2.2 Oxygen system – Figure 7
Acetylene system – Figure 8 (2)

2.3
- The nozzle hole must be clean.
- The nozzle hole must be square with the end of the nozzle. (2)

2.4 2.4.1 Backfiring occurs when the flame snaps back into the blowpipe or nozzle, producing a popping noise.

2.4.2 Flashback occurs when the flame snaps back into the hose and burns inside the torch.

(2 × 2) (4)

2.5 2.5.1 Undercut
2.5.2 Poor penetration
2.5.3 Poor fusion

(3 × 1) (3)

2.6
- All equipment must be cleaned after use.
- Cables must be rolled up.
- Repairs must be done immediately.
- Regular maintenance checks must be done.
- Ensure cylinders are closed.
- All tools must be stored correctly. (Any relevant answer) (Any 5 × 1) (5)

[20]

QUESTION 3

3.1 A scale of 1:100 means that 1 unit (mm/m) on the drawing represents 100 units (mm/m) on the job. (2)

3.2
- Moisture
- Indoor gas or fire fumes
- Diseases from humans and animals
- Tobacco smoke
- Formaldehyde from glues used in building materials
- Radon released from concrete structures
- Aerosols created from household products (Any 4 × 1) (4)

3.3
- The object's shape
- The weight of an object
- The size of an object
- The type of lifting equipment to be used
- The type of rigging equipment (Any relevant answer) (Any 3 × 1) (3)

3.4
- Ensure that you have the latest copy of the drawings
- Identify the scale of the drawings
- Identify the location
- Identify the component item number
- Highlight information on the drawing
- Check if markings correlate with the drawing

(Any relevant answer) (Any 3 × 1) (3)

3.5
- Eyes must be kept on the load.
- Equipment must always be inspected.
- Tyre pressure of the mobile crane must be checked.
- Safety devices must be checked.
- The crane hook must always be directly above the load to be hoisted.
- Ensure that the handbrake of the vehicle is always on during lifting.
- Examine the crane's ropes and slings.
- Handbrakes of the crane must be on when the load is hoisted/lowered.

(Any relevant answer) (Any 4 × 1) (4)

3.6
- It supports the strands by preventing them to jam.
- It lubricates the inside.
- It prevents corrosion.
- It provides a cushion effect to absorb shocks. (4 × 1) (4)

[20]

TOTAL: 100

Copyright reserved